WORKED SOLUTION

STEP III 1989, Question 4

Pure • Curve Sketching and Roots

STEP III 1989, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q4

The key idea

Splitting the fraction as a polynomial plus a proper fraction gives the oblique asymptote at once, and it also makes the derivative easy. The derivative turns out to be greater than 2 everywhere, so each of the three branches climbs from $-\infty$ to $+\infty$. Each equation in the second half is then "curve = some simpler curve" in disguise. Strict monotonicity of the difference on each branch then counts the crossings exactly.

Solution

The curve

Since $2x(x^2-5) = 2x(x^2-4) - 2x$,

$$y = 2x - \frac{2x}{x^2-4}.$$

The function is odd, so the curve has rotational symmetry of order 2 about the origin. It meets the axes only where $x(x^2-5)=0$, that is at $x=0$ and $x=\pm\sqrt{5}$.

Asymptotes. The denominator vanishes at $x=\pm 2$ (where the numerator is $\mp 4 \neq 0$), so $x=2$ and $x=-2$ are vertical asymptotes. As $|x|\to\infty$, $\frac{2x}{x^2-4}\to 0$, so $y=2x$ is an oblique asymptote. Also $y-2x = -\frac{2x}{x^2-4}$ is negative for large positive x and positive for large negative x. So the curve approaches $y=2x$ from below on the right and from above on the left.

Gradient. Differentiating the split form,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 - \frac{2(x^2-4) - 2x\cdot 2x}{(x^2-4)^2} = 2 + \frac{2(x^2+4)}{(x^2-4)^2} > 2.$$

So there are no stationary points, and y is strictly increasing on each of the intervals $x<-2$, $-2<x<2$ and $x>2$. Differentiating again gives $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\dfrac{4x(x^2+12)}{(x^2-4)^3}$, so the only point of inflection is the origin.

Tangent at the origin. At $x = 0$ the gradient is $2 + \frac{2\cdot 4}{16} = \frac{5}{2}$, so the tangent there is

$$y = \tfrac{5}{2}x.$$

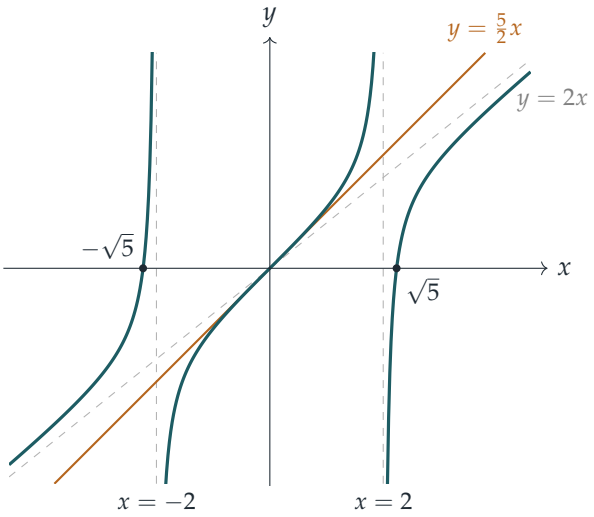
Behaviour near $x = \pm 2$. Near $x = 2$ the term $-\frac{2x}{x^2-4}$ dominates. It tends to $-\infty$ as $x \to 2^+$ and to $+\infty$ as $x \to 2^-$, and by symmetry it tends to $+\infty$ as $x \to -2^-$ and to $-\infty$ as $x \to -2^+$. So:

- for $x < -2$, y rises from $-\infty$ to $+\infty$, crossing the axis at $-\sqrt{5}$;
- for $-2 < x < 2$, y rises from $-\infty$ to $+\infty$ through the origin;
- for $x > 2$, y rises from $-\infty$ to $+\infty$, crossing the axis at $\sqrt{5}$.

One more fact will be useful. For $0 < x < 2$,

$$y - \tfrac{5}{2}x = \frac{x\left[4(x^2-5) - 5(x^2-4)\right]}{2(x^2-4)} = \frac{-x^3}{2(x^2-4)} > 0,$$

so on this stretch the curve lies above its tangent at the origin (and, by symmetry, below it for $-2 < x < 0$).



Part (i)

Neither $x = 0$ (LHS 0, RHS 8) nor $x = \pm 2$ (LHS -16, RHS 0) is a root. So we may divide by $2x(x^2-4)$, and the equation becomes

$$\frac{2x(x^2-5)}{x^2-4} = \frac{5x-2}{2x} = \frac{5}{2} - \frac{1}{x}.$$

The roots are therefore the x-coordinates of the intersections of the curve with the hyperbola $H\colon y = \frac{5}{2} - \frac{1}{x}$. We go branch by branch.

- $x < -2$: here H has gradient $\frac{1}{x^2} < \frac{1}{4}$, while the curve has gradient > 2. So the difference (curve $-$ H) is strictly increasing. It tends to $-\infty$ as $x \to -\infty$ and to $+\infty$ as $x \to -2^-$, because H stays between $\frac{5}{2}$ and 3. Exactly one root.
- $-2 < x < 0$: the curve has $y < 0$ (all three factors x, $x^2 - 5$, $x^2 - 4$ are negative), but H has $y > \frac{5}{2}$. No roots.
- $0 < x < 2$: by the fact above, $y > \frac{5}{2}x$. Also

$$\tfrac{5}{2}x - \left(\tfrac{5}{2} - \tfrac{1}{x}\right) = \frac{5x^2 - 5x + 2}{2x} > 0,$$

  since $5x^2 - 5x + 2$ has discriminant $25 - 40 < 0$. So the curve lies strictly above H. No roots.
- $x > 2$: as in the first case, the difference is strictly increasing (gradient $> 2 - \frac{1}{4}$). It goes from $-\infty$ to $+\infty$, because H stays between 2 and $\frac{5}{2}$. Exactly one root.

So equation (i) has exactly **two** real roots, one less than -2 and one greater than 2.

Ask yourself How can the equation in (ii) be read off the curve when it has squares on both sides?

Divide by $(x^2 - 4)^2$. The left side becomes y^2, so the equation says $y^2 = x^2 + 1$.

Part (ii)

Again $x = \pm 2$ is not a root (LHS 16, RHS 0). So the equation is $y^2 = x^2 + 1$, where y is the curve's ordinate. The roots are the intersections of the curve with the two branches $y = \pm\sqrt{x^2 + 1}$ of the rectangular hyperbola $y^2 - x^2 = 1$.

Upper branch. Let $f(x) = y - \sqrt{x^2 + 1}$. Then

$$f'(x) = \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{x}{\sqrt{x^2 + 1}} > 2 - 1 > 0,$$

so f is strictly increasing on each of the three intervals. On each one, y runs from $-\infty$ to $+\infty$ while $\sqrt{x^2 + 1}$ stays finite (it grows only like $|x|$, against $2x$ as $x \to +\infty$; and as $x \to -\infty$ both terms of f are negative). So f goes from $-\infty$ to $+\infty$, and there is exactly one root on each interval: three in all.

Lower branch. The map $(x, y) \mapsto (-x, -y)$ sends the curve to itself and the lower branch to the upper one. So there are also exactly three intersections with the lower branch.

The two sets of roots don't overlap: at a common x the curve has a single y, which can't be both positive and negative. So equation (ii) has exactly **six** real roots. (As a check, it is a polynomial of degree 6, namely $3x^6 - 33x^4 + 92x^2 - 16 = 0$, so all its roots are real.)

Ask yourself The variable in (iii) is z, not x, and z might be negative. How do we bring the curve in?

For $z > 0$ we put $z = x^2$ with $x > 0$. We deal with $z \leq 0$ separately and directly.

Part (iii)

The left side is never negative, so any root must have $z + 1 \geq 0$, that is $z \geq -1$. Also $z = 0$ is not a root (LHS 0, RHS 16), and neither is $z = 4$ (LHS 64, RHS 0).

Roots with $z > 0$. Put $z = x^2$ with $x > 0$, so that each positive root z corresponds to exactly one $x > 0$. The equation becomes $4x^4(x^2 - 5)^2 = (x^2 - 4)^2(x^2 + 1)$. Dividing by $x^2(x^2 - 4)^2$ gives

$$y^2 = 1 + \frac{1}{x^2}, \qquad \text{that is} \qquad y = \pm s(x), \quad s(x) = \sqrt{1 + x^{-2}}.$$

Here s decreases from $+\infty$ (as $x \to 0^+$) to 1 (as $x \to \infty$), and $s'(x) = -\dfrac{1}{x^3\sqrt{1 + x^{-2}}}$, so $|s'(x)| < \frac{1}{x^3}$.

- $y = s(x)$: the function $y - s$ has derivative $y' - s' > 2$. On $(0, 2)$ it runs from $-\infty$ (as $y \to 0$ and $s \to \infty$) to $+\infty$. On $(2, \infty)$ it also runs from $-\infty$ to $+\infty$. One root on each, so two roots.
- $y = -s(x)$ on $(0, 2)$: the curve has $y > 0 > -s$ there. No roots.
- $y = -s(x)$ on $(2, \infty)$: the function $y + s$ has derivative $y' + s' > 2 - \frac{1}{8} > 0$ and runs from $-\infty$ to $+\infty$. One root.

So there are exactly three positive roots z.

Roots with $-1 \leq z < 0$. Write $z = -t$ with $0 < t \leq 1$. The equation becomes

$$4t^2(t + 5)^2 = (t + 4)^2(1 - t).$$

On $(0, 1]$ the left side is strictly increasing, from 0 (as $t \to 0$) to 144. The right side is strictly decreasing, from 16 to 0. Their difference is strictly increasing and changes sign, so there is exactly one root.

Hence equation (iii) has exactly **four** real roots: three positive and one in $(-1, 0)$. Expanded, it is the quartic $4z^4 - 41z^3 + 107z^2 - 8z - 16 = 0$, so all four of its roots are real.

Checks

- At $x = 1000$, $y - 2x \approx -0.002$, which confirms the approach to $y = 2x$ from below.
- Numerically, (i) has roots $x \approx -2.143$ and 2.401, one on each outer branch as predicted.

- (ii) has roots $x \approx \pm 0.431,\ \pm 2.155,\ \pm 2.484$, one per branch of the curve for each branch of the hyperbola.
- (iii) has roots $z \approx -0.332,\ 0.473,\ 4.799,\ 5.310$. The last three give $x = \sqrt{z} \approx 0.688,\ 2.191,\ 2.304$, with curve values $y \approx 1.77,\ -1.10,\ 1.09$, matching the case analysis.

Common slips

- Dividing by x or $x^2 - 4$ without first checking that $x = 0$ and $x = \pm 2$ aren't roots.
- In (ii), counting only the intersections with $y = +\sqrt{x^2 + 1}$ and forgetting the lower branch, which gives 3 instead of 6.
- In (iii), substituting $z = x^2$ and forgetting that z may be negative, which misses the root in $(-1, 0)$.

Beyond STEP

The counting arguments here rest on the intermediate value theorem (a continuous function that changes sign has a zero) together with strict monotonicity (it can have at most one). Making these two facts rigorous is one of the first jobs of a university analysis course. Abbott's *Understanding Analysis* treats both carefully and readably. Descartes' rule of signs and Sturm sequences are algebraic ways to count real roots of polynomials, and they make an interesting comparison with the geometric method used here.

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